

# Anabel Yong

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## Summary

MSc in Computational Statistics & Machine Learning from UCL with interests in Gaussian Processes, Bayesian Optimisation, and kernel-based methods. Interested in machine learning for drug discovery and learning biological data.

## Papers

- **Yong, A., Tripp, A., Hosseini-Gerami, L., Paige, B. (2025). Bayesian Optimization for Molecules should be Pareto-Aware. (Accepted as Poster, NeurIPS 2025 AI for Science Workshop.) Link: [arXiv](#)**

## Education

**MSc Comp. Statistics & Machine Learning, UCL (Distinction, Dean's List)** Sept 2023 - Sept 2024  
**Modules:** Probabilistic & Unsupervised Learning (Gatsby PhD module) (91%), Machine Learning Seminar (Marc Deisenroth- DeepMind)(76%), Numerical Optimization (80%), Statistical Natural Language Processing

**BSc Maths & Biology, University of Edinburgh (First Class Honours)** October 2019 - June 2023  
**Modules:** Bioinformatics (95%), Machine Learning (80%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (76%), Mathematics for Natural Sciences (80%)

## Work Experience

- **Research Intern, National University of Singapore (PI: Jonathan Scarlett)** May 2024 - Present
  - Designing an batched Bayesian optimization algorithm to promote structural diversity in small molecule optimization.
- **Machine Learning Research Intern, Ignotalabs.AI (PIs: Austin Tripp, Cambridge MLG Group & Brooks Paige, UCL)** June 2024 – Sept 2024
  - Developed a multi-objective Bayesian optimization algorithm using Tanimoto/MinMax Kernels for small molecule optimization.
- **Research Intern, Kustatscher Lab, University of Edinburgh** Sept 2022 - July 2023
  - Benchmarked a data pre-processing pipeline for proteomics, pgFDR), to enhance protein detection accuracy.
  - Aided in database curation of human proteins and microproteins of ProteomeHD2
- **Research Intern, ZiHeng Yang Lab, UCL** March 2022 - May 2022
  - Developed Bayesian MCMC algorithms to model ABO blood type frequencies, contributing to population genetics and personalised medicine approaches.
- **Research Intern, Grima Lab, University of Edinburgh** June 2020 - Jan 2022
  - Implemented stochastic and deterministic models to predict mRNA decay kinetics, aiding in the understanding of gene regulation mechanisms and supporting drug efficacy studies in preclinical research.

## Skills

**Programming:** Python, R, MATLAB, git, Linux, Google Cloud Platform (GCP)

**Machine Learning:** PyTorch, JAX, BoTorch, GPyTorch

**Bioinformatics/Cheminformatics:** RDKit, PyMOL, UniProt, MaxQuant

## References (available upon request)

- **Brooks Paige, Associate Professor in Machine Learning, University College London:** [b.paige@ucl.ac.uk](mailto:b.paige@ucl.ac.uk)
- **Jonathan Scarlett, Associate Professor in Dept. of Computer Science & Mathematics, National University of Singapore:** [scarlett@comp.nus.edu.sg](mailto:scarlett@comp.nus.edu.sg)
- **Layla Hosseini-Gerami, Chief Data Science Officer, Ignotalabs.ai:** [layla.gerami@ignotalabs.ai](mailto:layla.gerami@ignotalabs.ai)
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** [austin.james.tripp@gmail.com](mailto:austin.james.tripp@gmail.com)